# Ziang (Jay) Zhang

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**Interactive AI Portfolio – Ask Me Anything:** jayziangzhang.github.io/askme/

AI and Digital Transformation Engineer with experience in data analytics, predictive modeling, industrial automation, and GenAI solutions across manufacturing and corporate environments.

Built production-grade IT/OT integration systems, edge data platforms, and Retrieval-Augmented Generation (RAG) applications. Experienced in full-stack development, enterprise AI adoption, and large-scale digital transformation initiatives.

- **Programming**
  Python • SQL • JavaScript • TypeScript
- **AI & Machine Learning**
  GenAI • LLMs • RAG • Agentic AI • Prompt Engineering
- **Data & Analytics**
  Data Analytics • Statistical Analysis • Predictive Modeling
- **Backend & Infrastructure**
  Node • REST APIs • Redis • RabbitMQ • Docker
- **Industrial & Edge Computing**
  PLC Integration • IT/OT Integration • Edge Computing
- **Tools & Methodologies**
  Git • Agile • Scrum

## EDUCATION

**University of Toronto**
Master of Management Analytics, 2024
BASc, Electrical Engineering, 2021 | Minor in Engineering Business | Certificate in Artificial Intelligence

## PROFESSIONAL EXPERIENCE

**Technical Consultant – Factory Automation & Digitalization**
***Siemens* | *Beijing, China* | *2021–2023***

- Delivered industrial digitalization and IT/OT integration solutions for 30+ manufacturing and agricultural facilities, enabling customers to transition from manual operations to digital workflows.
- Developed a high-frequency industrial edge data acquisition platform for Great Wall Motors using HTTP, Redis, and RabbitMQ, enabling real-time ingestion and processing of manufacturing equipment data for enterprise system integration.
- Built energy management applications for Swire Coca-Cola, integrating PLC and utility meter data to calculate production-line energy KPIs and identify abnormal energy consumption patterns.
- Designed middleware solutions connecting shop-floor equipment and MES systems, implementing device monitoring, heartbeat detection, automated data validation, and production data synchronization.

**GenAI Consultant (Practicum)**
***Manulife* | *Toronto, Canada* | *2024***

- Developed a proof-of-concept Retrieval-Augmented Generation (RAG) HR assistant enabling employees to query internal policies using natural language.
- Designed and deployed an AI-powered policy authoring platform for HR teams, automating policy generation while enforcing corporate writing and formatting standards.
- Presented GenAI solutions to business stakeholders, demonstrating practical applications of enterprise AI for HR operations.

**Founder & Full-Stack Developer – IDP Plus | www.idp-plus.com**

- Built a SaaS platform for international driver's license translation and generation, processing 500+ customer orders through a fully automated workflow that reduced manual document preparation time by 95%.
- Developed the full technology stack, including frontend, backend APIs, OCR-based document validation, automated PDF generation, and payment processing workflows.
- Collaborated with external partners on printing, distribution, and marketing while independently managing deployment, website maintenance, and platform operations.